

9 (short)

(a) Figure 7 shows a sketch of a spanner which is to be manufactured in carbon steel, with an expected batch size of 5000. Selected dimensions of the spanner with the required precisions are shown in Fig. 7. The thickness is  $4 \pm 0.1$  mm. Use the process attribute charts in the Materials Databook to choose between sand casting and forging, basing your selection on the following criteria: mass, tolerance, and economic batch size. Explain why the manufacturing process also involves machining. [6]

(b) The manufacturing process also involves a final heat treatment. State which hardening mechanism is exploited via heat treatment of carbon steel, and briefly summarise the microstructural mechanism. State with brief reasons whether the heat treatment is also expected to affect the Young's modulus and the fracture toughness of the steel. [4]

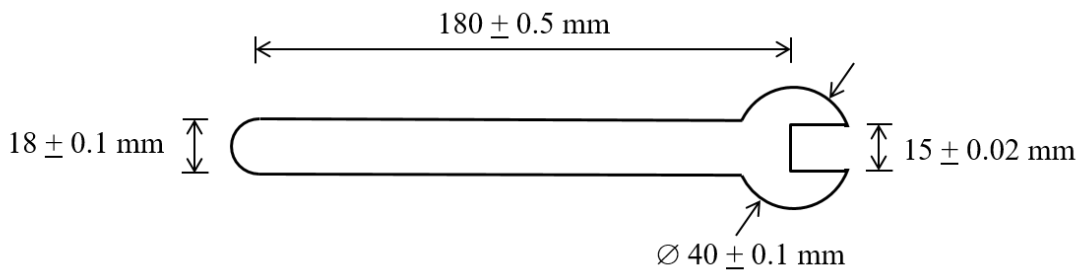


Fig. 7